



Gastrointestinal Medicine in the Athlete

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Gastrointestinal (GI) conditions in athletes and active individuals are common and are seen frequently by sports medicine care providers. Although there are many well-defined benefits that result from high-level physical activity, exercise places significant stress on the GI tract, which in turn may result in a number of characteristic GI disorders. Collectively, GI problems are perhaps the most common cause of underperformance in endurance sports and may also impact on subsequent recovery. The appreciation of this relationship is important for both care providers and athletes who must team together to minimize distressing GI symptoms and to prevent subsequent impairment in performance.

GI disorders in athletes often mimic those seen in the general population but may also present as unique entities seen primarily in those who exercise at a high level. This chapter will address a number of these common conditions, with a focus on their recognition and appropriate management in athletes.

PATHOPHYSIOLOGY

Frequent, high-intensity exercise may cause a number of undesirable GI symptoms, such as heartburn, chest pain, bloating, belching, nausea, vomiting, abdominal cramps, frequent urge to defecate, diarrhea, and GI bleeding. Historically, these symptoms have been reported by up to 50% of endurance athletes and are more common in women and younger athletes.¹ Body position and movement appear to play a role because the forward positioning of cyclists appears to protect them from GI tract issues as opposed to the higher-impact, jostling abdominal movements of runners.² Other suggested mechanisms for increased GI distress include a number of neuro-immuno-endocrine adaptations to exercise. Increased catecholamines and several GI peptides including gastrin, motilin, secretin, peptide histidine-methionine, and vasoactive intestinal peptide result in increased gut transit time with negative effects on digestion and absorption.³⁻⁶

Gastrointestinal Tract Ischemia

GI tract ischemia secondary to attenuated GI blood flow^{4,7} is a well-defined exercise adaptation that results from the redirection of blood flow to active muscles and the pulmonary vasculature. Vigorous exertion, hyperthermia, dehydration, hypoglycemia, mental stress, and fatigue all increase sympathetic nervous system activity, which in turn shunts blood away from the GI viscera to provide increased blood and oxygen supply to working muscles. Exercising moderately at up to 70% of maximum oxygen uptake

($\text{VO}_{2\text{max}}$) reduces blood flow to the GI tract by 60% to 70%, and higher-intensity exercise may result in GI blood flow reductions exceeding 80%.⁵ Subsequent impairment in oxygen delivery to GI tract structures may result in mucosal injury, increased gut mucosal permeability, increased risk for occult blood loss, translocation of protective bacterial flora, and generation of endotoxins.^{4,8} These changes contribute to ischemia-associated GI symptoms, including nausea, vomiting, abdominal pain, and bloody diarrhea.⁹ It has been postulated that GI complaints relating to gut ischemia are the result of a reperfusion injury after exercise has ceased. Reperfusion after ischemia may result in a number of chemical and vascular changes that result in “leaky mucosa.” When this occurs, the GI tract partially loses its barrier ability to protect itself from inherently irritating intraluminal substances such as endotoxins, food antigens, digestive enzymes, and bile.¹⁰

Impact of Nutrition and Hydration

The overall fluid status of athletes has been shown to play an important role in the relative GI toxicity of high-level activity. Dehydration contributes significantly to GI tract dysfunction because it further accentuates the intrinsic blood flow changes associated with exercise, as noted previously. Research has shown that 80% of runners who lose at least 4% of their body weight from fluid losses during exercise experience lower GI tract symptoms.² Exercise-related fluid and electrolyte shifts result in intracellular electrolyte imbalances and cell dysfunction, which may lead to colonic smooth muscle and mucosal irritation.¹¹ In runners with restricted water intake, GI symptoms are more likely to arise due to increases in upper GI tract and intestinal permeability relative to resting individuals.¹² In a study of nutritional intake patterns in endurance athletes,¹³ it was shown that high rates of carbohydrate intake (up to 90 g/h) were associated with an increased risk for nausea and flatulence. In contrast, ingestion of a lower-carbohydrate liquid meal before exercise is generally well tolerated, may help to maintain GI tract perfusion,¹⁴ and may prevent gut ischemia related to decreased splanchnic blood flow.

UPPER GASTROINTESTINAL TRACT CONDITIONS

Upper GI tract complaints are experienced by 30% to 70% of athletes,^{10,15,16} with the prevailing conditions being gastroesophageal

Abstract

Gastrointestinal (GI) conditions in athletes and active individuals are common because exercise places significant stress on the GI tract and results in a number of characteristic GI disorders. Collectively, GI problems are perhaps the most common cause of underperformance in endurance sports and may also impact on subsequent recovery. The appreciation of this relationship is important to minimize distressing GI symptoms and to prevent subsequent impairment in performance. GI disorders in athletes often mimic those seen in the general population but may also present as unique entities seen primarily in those who exercise at a high level. This chapter will address a number of these common conditions, with a focus on their recognition and appropriate management in athletes.

Keywords

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reflux disease (GERD) and dyspepsia.^{17,18} When assessing the pathophysiology behind the high rate of upper GI tract symptoms in athletes, three primary mechanisms have been postulated: (1) mechanical forces, (2) alterations in GI tract blood flow, as noted previously, and (3) neuroendocrine changes that influence overall GI tract function. Mechanical forces include increased abdominal pressure from straining in high-exertion activities such as weightlifting, mechanical trauma to the GI tract from repetitive motion of the abdomen during exercise, and the impact of body position during exercise such as the flexed/compressed position of cyclists.¹⁹ GI mucosal activity and alterations in absorption may result from mechanical factors or neuroendocrine changes. Neural activity changes, as referenced previously, have all been documented to negatively influence GI tract function in athletes.⁵

Exercise also greatly affects esophageal motility. Relaxation of the lower esophageal sphincter (LES), increased pressure gradient between the stomach and esophagus, and decreased esophageal clearance of food all have been associated with high-level physical exertion, and all appear to be potentially important in the development of GERD symptoms.²⁰ As exercise intensity reaches peak levels exceeding 90% of $\text{VO}_{2\text{max}}$, the frequency and duration of esophageal reflux episodes increase.^{21,22} Exercise type also influences esophageal motility. Running has been found to decrease upper GI tract motility more than cycling,²³ and the highest rate of heartburn and esophageal reflux has been seen in weightlifters, with correspondingly lower rates found in runners and cyclists.²⁴

Gastroesophageal Reflux Disease

GERD is the most common cause of upper GI tract symptoms in athletes and results from the irritating effects of acidic gastric secretions as they reflux up into the esophagus due to loss of competence of the LES. The incidence of GERD increases with intensity of exercise, is more common in endurance sports, and is exacerbated by postprandial exercise. Runners demonstrate a threefold increase in GERD symptoms when running 45 minutes after completing a meal as compared with running in a fasting state.²⁴ Activities with large increases in intra-abdominal pressure, most notably weightlifting and cycling when in the “aero” position, contribute to higher rates of symptomatic GERD as the LES is overwhelmed by the external pressure applied to the stomach.

Cardinal features of GERD include heartburn, retrosternal burning, and regurgitation with a sense of refluxed gastric contents arising into the mouth or hypopharynx.²⁵ The majority of athletes with true exertional GERD actually have GERD at rest as well²⁶; thus the greatest risk factor for GERD symptoms during exercise is the presence of GERD symptoms at rest.

Gastroesophageal Reflux Disease Pathophysiology

Although precise GERD mechanisms are not well defined, suggested mechanisms include inappropriate relaxation of the LES,⁴ gastric dysmotility, enhanced pressure gradient between the stomach and esophagus in sports such as football, weightlifting, and cycling, gastric distension, delayed gastric emptying, and increased mechanical stress by bouncing of organs.⁴

Normal peristaltic motions of the esophagus, which aid in ensuring that the acidic contents of the stomach remain in the stomach and do not migrate superiorly into the neutral environment of the esophagus, decrease in the setting of higher levels of activity. Thus any food or fluid intake that fails to pass through the stomach prior to exercise predisposes the athlete to reflux. Food choice is important because high-fat foods (fried foods, creamy sauces, and gravy), coffee, caffeine, chocolate, peppermint, alcohol, acidic foods such as tomatoes and onions, and tobacco products are well known to decrease LES pressure. In exercising individuals, the widespread use of nonsteroidal antiinflammatory drugs (NSAIDs) may also impact negatively on the proximal GI tract due to inherent GI tract irritation from suppression of protective prostaglandins.

Gastroesophageal Reflux Disease, Asthma, and Other Related Conditions

GERD may present with a number of less typical symptoms, including cough, sore throat, hoarseness, asthma, bronchitis, recurrent pneumonia, intermittent choking, or chest pain. Athletes presenting with any of these symptoms or complaints should be considered potential GERD sufferers. Many of these individuals have “silent” reflux that fails to generate classic GI manifestations. GERD can provoke or exacerbate asthma by either silent aspiration into the tracheobronchial tree or by stimulating an acid-mediated esophagobronchial reflex that provokes dyspnea and wheezing.²⁷ It is estimated that up to 90% of asthmatics have GERD and that up to 40% of asthmatics have esophagitis.²⁸ However, to date only those with nocturnal GERD symptoms have demonstrated improvements in their asthma from aggressive GERD treatment.²⁹ It is always important to consider GERD as the potential primary pathology in an athlete who has new or unusual asthmatic symptoms.

Several clinical variants of GERD are also recognized. Laryngopharyngeal reflux disease (LPRD) is typically manifest by hoarseness, voice change, and chronic cough. It is believed to result from an esophageal reflux mechanism with secondary inflammation and damage of the proximal airway and laryngeal area. Duodenogastroesophageal reflux (DGER) results from reflux of duodenal, rather than gastric, contents, including biliary secretions, pancreatic enzymes, and bicarbonate. DGER is often responsible for refractory GERD symptoms despite appropriate management with maximal dose medications.³⁰ Finally, a number of proposed extraesophageal manifestations of GERD include recurrent pharyngitis, otitis media, sinusitis, and pulmonary fibrosis.

Gastroesophageal Reflux Disease Treatment

The importance of proper GERD management cannot be overstated, not only to prevent acute symptoms and their impact on sport performance but also to decrease the risk of long-term complications including peptic strictures and Barrett esophagus with its potential for malignant transformation. Successful GERD management includes lifestyle modifications, medications, or both. Lifestyle modifications should always be the first line treatment in athletes who have GERD. These include sleeping on two pillows or elevating the head of the bed to enhance gravity-associated

esophageal clearance, avoidance of laying down to sleep within 4 hours of the evening meal, avoiding postprandial exercise, and limiting consumption of foods known to relax the LES, as noted previously. Athletes should also avoid solid food, high carbohydrate drinks, caffeine and alcohol prior to strenuous activity.¹⁹ Rather, they should consume liquid meals with lower carbohydrate content and ensure adequate hydration. Sports drinks containing up to a maximum of 10% glucose are believed to be the optimal calorie-containing beverages for training because of their low tendency to cause upper GI tract symptoms.²⁶

If these interventions fail to improve symptoms, pharmacologic therapy should be considered. The two primary classes of medications used to treat GERD include proton pump inhibitors (PPIs; e.g., omeprazole, lansoprazole, pantoprazole, esomeprazole) and H₂-receptor antagonists (e.g., ranitidine, famotidine). Initially, a 2-week trial of a once-daily PPI such as 20-mg omeprazole or 30-mg lansoprazole, taken prior to a meal, should be instituted. If once-daily therapy is inadequate, PPI dosing should be doubled. If initially effective, completion of an 8-week course is generally appropriate and effective for long-term benefit, particularly if combined with lifestyle measures. For GERD symptoms limited to discreet competitive times, it has been shown that markers of exercise-induced gastroesophageal reflux—percentage reflux time and number of reflux episodes—can be effectively reduced by pretreatment with an H₂-receptor antagonist (i.e., 300 mg ranitidine) 1 hour prior to activity.³¹ Rapid onset of benefit with H₂ blockers makes them the preferred therapies for such acute indications.

If symptoms recur within 3 months, additional evaluation to confirm the diagnosis is generally warranted to determine the appropriateness of ongoing therapy. Upper endoscopy is the evaluation tool of choice for diagnosis confirmation and is also indicated for more concerning clinical presentations, including failure of empiric therapy, need for continuous therapy, presence of atypical symptoms such as odynophagia, dysphagia, weight loss, bleeding, or anemia, or if symptoms are long-standing. Additional workup should include *Helicobacter pylori* testing to exclude its potential role in non-GERD syndromes such as peptic ulcer disease and nonulcer dyspepsia. Confirmation of *H. pylori* infection is most effectively achieved via gastric mucosal biopsy at the time of upper endoscopy or via serologic, breath, or stool antigen testing. If such testing is positive, a 2-week triple-drug antibiotic/acid suppressive combination treatment course should be pursued. Such combination therapies include PPI + clarithromycin + amoxicillin or metronidazole, or PPI + a bismuth compound + metronidazole + a tetracycline.

Functional Heartburn

Studies reveal that up to 60% of individuals presenting with signs and symptoms compatible with GERD but who fail aggressive PPI therapy have a syndrome referred to as “functional heartburn.” This disorder is also known as “esophageal hypersensitivity” and bears resemblance to other common functional GI disorders that include a pattern of visceral hyperalgesia such as irritable bowel syndrome (IBS).¹⁷ The Rome III diagnostic criteria for functional heartburn include burning retrosternal discomfort or pain, absence of evidence that GERD is the cause

of the symptom, absence of an esophageal motility disorder, and symptom onset of at least 6 months.¹⁸ As such, this diagnosis can be made only following formal referral to a gastroenterologist with completion of an appropriate diagnostic workup.

Management is challenging and many sufferers continue to have symptoms despite therapy. Biofeedback to modify nervous system input to the GI tract is the treatment of choice. Tricyclic antidepressants and prokinetic agents also have potential value, but their significant side effect profiles are generally incompatible with sport participation.

Dyspepsia

Dyspepsia refers to a spectrum of vague GI symptoms, including gnawing or burning epigastric pain, nausea, vomiting, eructation, bloating, indigestion, generalized abdominal discomfort, and early satiety. The three most common causes of dyspepsia in athletes include GERD, gastritis, and peptic ulcer disease. In addition, many drugs are associated with dyspepsia, including NSAIDs, antibiotics, and estrogens. In fact, most drugs have a potential to cause dyspepsia in at least some patients.

Regardless of the cause of dyspepsia, the common pathologic feature is mucosal inflammation and damage of the upper GI tract. Gut mucosal injury in athletes may be multifactorial from mucosal ischemia, dehydration, stress of competition, use of NSAIDs or a variety of medications, alcohol, caffeine products, or dietary supplements. These same factors increase the risk for upper GI tract bleeding and ulcer disease in athletes. Upper GI evaluations in professional long-distance runners revealed at least one GI mucosal lesion in approximately 90% of runners even in the absence of symptoms.³²

Dyspepsia management focuses on the limitation or elimination of the provocative factors mentioned previously. Should those interventions provide inadequate benefit, acid suppressive therapy with H₂ blockers or PPIs is the pharmacologic treatment of choice. For those individuals who suffer from chronic dyspepsia or who have had frank GI bleeding with heavy exercise, consistent use of a H₂ blocker or PPI is prudent to prevent symptoms or complications from upper GI tract damage. Data from ultramarathon runners have shown that regular use of a PPI significantly decreased their incidence of GI tract bleeding.³³

Impact of Aspirin and Nonsteroidal Antiinflammatory Drugs

Aspirin and NSAID use places the GI tract at risk due to the blockade of cyclooxygenase and the subsequent loss of production of protective gastric prostaglandins. Although clearly linked to a risk of gastritis and peptic ulcer disease with prolonged use, NSAIDs have yet to be shown to correlate with risk for GI bleeding in endurance athletes.³⁴ Whenever feasible, it is advisable to have athletes minimize the use of NSAIDs; however, the need for pain and inflammation management in highly active individuals makes complete avoidance of NSAIDs unrealistic. For those who require NSAIDs, it is prudent to add a daily PPI for GI tract protection, particularly if they have a previous history of ulceration or bleeding. Special note should be made of the NSAID ketorolac. Although it possesses analgesic properties equivalent to opioid therapy, ketorolac carries a high risk of GI

bleeding with long-term use. Present dosing guidelines mandate discontinuation of ketorolac therapy after a maximum of 5 days of use.

Pill Esophagitis

Pill esophagitis results from the local irritating effects of certain medications if they remain in the esophagus for a prolonged amount of time. In athletes, NSAIDs are the classic family of medications with greatest risk. The tetracycline antibiotic family, often used in athletes chronically for acne or in shorter-term courses for tick-borne illness, and the antimalarial mefloquine are notoriously problematic for this condition as well. Prevention is best achieved by taking medication in the morning with food or a moderate volume of liquid, remaining upright for at least 30 minutes after pill ingestion, and avoiding bedtime dosing. It is always important to educate athletes about a medication's potential risk for GI toxicity as well.

Dysphagia

Dysphagia is a condition characterized by difficulties with swallowing, which are typically driven by alterations in either nerve or muscle function. Sport-associated dysphagia is most commonly caused by the irritative effects of GERD, although any intrinsic esophageal disorder may be responsible. Clinically, dysphagia is divided into two subtypes based on anatomic location and pathologic mechanism. Oropharyngeal dysphagia (OD) typically presents with greater difficulty swallowing liquids than solids and is characterized by repeated attempts to swallow due to a sensation of choking or incomplete swallowing. OD occurs within the first second of the initiation of swallowing and presents with discomfort or pain that localizes to the neck. Other characteristics may include a change in dietary habits, dehydration, regurgitation, or unexplained weight loss. In contrast, esophageal dysphagia (ED) presents with difficulty swallowing both solids and liquids, occurs several seconds after the initiation of swallowing, and is characterized by discomfort or pain that localizes to the substernal area. Dysphagia management focuses on addressing presumptive GERD or identifying and managing other underlying conditions.

Nausea and Vomiting

Nausea and vomiting may be experienced by athletes in association with strenuous or prolonged exercise.¹⁵ Although the etiology is not always clear and thus may create a challenge for prevention and management, nausea and vomiting are most often associated with postprandial exercise. Reflux of stomach contents coupled with changes in esophageal motility, sphincter tone, and GI ischemia contributes to the increased incidence of nausea and vomiting associated with exercise. Nevertheless, athletes may have these manifestations even when relatively inactive, a phenomenon supported by research in runners who actually demonstrated a higher incidence of bloating and vomiting when not training.³⁵ Although the reasoning is unclear, this is likely a function of the chronic irritative effects of running on the GI tract.

Prevention of nausea and vomiting focuses on prudent dietary and fluid choices to minimize gastric irritation or altered GI absorption. In keeping with general intake guidelines, for events

lasting less than 1 hour, optimal energy supplementation to minimize GI effects is provided by sports drinks with carbohydrate concentrations of 6% to 8%. For longer duration activities, carbohydrate intake of 30 to 60 g/h should be well tolerated and effective.³⁶ Athletes should also avoid solid food within 3 hours of an athletic event. Finally, because NSAID use has been shown to increase nausea and vomiting in endurance runners,³⁷ caution should be exercised in their use around the time of athletic participation even for activities of shorter duration.

The primary treatment for nausea and vomiting associated with exercise is the prompt cessation of physical activity. However, in athletes with protracted nausea and vomiting, evaluation for complications of dehydration and electrolyte disturbances should be pursued. In addition, other causes such as peptic ulcer disease and biliary pain should be considered. If nausea is associated with prominent heartburn during activity, a trial of H2 blocker or PPI may be warranted. Studies have shown that H2 blockers and PPIs, as well as anti-nausea therapies such as ondansetron, can be helpful for exercise-related nausea, and the sports medicine provider should have a low threshold to treat affected athletes when they are participating in activities that are likely to provoke symptoms.³⁸

LOWER GASTROINTESTINAL TRACT CONDITIONS

As with upper GI disorders, issues relating to the small intestines and colon are common in sport, with nearly two-thirds of endurance athletes reporting lower GI symptoms at some point in their training or competition.³⁹ Conditions such as diarrhea and rectal bleeding are often directly related to intensity of activity, whereas functional disorders, as seen frequently in the general population, are often exacerbated by the physical demands of exercise. Although the typical athlete presents with the conditions discussed next, the differential is broad and must be considered individually for each case (Table 15.1).

Diarrhea

Diarrhea results from excessive fluid content of stool that overwhelms the colon's absorptive ability and results in loose or frankly watery bowel movements. Diarrhea is a common concern in athletes, with approximately 25% of marathon runners reporting diarrhea and more than half reporting fecal urgency with running. As expected, the severity of diarrhea is directly proportional to training mileage or training intensity, with women affected more commonly than men.^{11,39} Although generally benign, prominent diarrhea in athletes may be associated with dehydration and subsequent risk for heat injury, rhabdomyolysis, and acute tubular necrosis.

The etiology of exercise-related diarrhea is often multifactorial. Increased sympathetic nervous system stimulation with exercise decreases gut tone and resistance, accelerates transit of colonic contents into the rectum, and increases the urge to defecate.⁴⁰ Gastroenteropancreatic hormone secretion of gastrin and motilin with high-level exercise may further contribute to the development of diarrhea.⁴¹ Nutritional factors may cause diarrhea relative to their ability to increase stool water content. More highly

TABLE 15.1 Differential Diagnosis of Lower Gastrointestinal Symptoms in Athletes**Nutritional Factors**

- Dietary fiber
- High glycemic index foods
- High carbohydrate concentration sports drinks or gels
- Food allergies and intolerances

**Irritable Bowel Syndrome
Malabsorption Syndromes**

- Lactose intolerance
- Celiac disease

Medications, Herbals, Supplements

- Laxatives and cathartics
- H₂ blockers and antacids
- Nonabsorbable sugars (e.g., mannitol, sorbitol)
- Antibiotics
- Caffeine
- Alcohol
- NSAIDs

Infectious Diarrhea

- Bacterial
- Parasitic
- Viral

Toxic Diarrhea

- Pseudomembranous enterocolitis (*Clostridium difficile*)
- Staphylococcal toxin

Inflammatory Bowel Disease

- Crohn disease
- Ulcerative colitis

Ischemic Bowel Disease

- Acute arterial mesenteric infarction
- Chronic mesenteric ischemia
- Mesenteric venous thrombosis
- Colonic ischemia

**Hepatic, Biliary, and Pancreatic Diseases
Lower GI Tract Lesions**

- Ulcerations
- Polyps
- Arteriovenous malformations
- Anorectal hemorrhoids, fissures, or fistulae

Colon Cancer and Other Neoplastic Processes

GI, Gastrointestinal; NSAIDs, nonsteroidal antiinflammatory drugs.
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Curr Sports Med Rep. 2009;8(2):85–91.

concentrated fluids (>7% to 10%) present a solute load to the GI tract, which produces an osmotic drive for fluid to remain in the GI tract rather than being absorbed. This shift toward fluid deposition into the gut produces a “dumping syndrome” that overwhelms the colon’s absorptive abilities and results in diarrhea. For endurance athletes, it has been noted that when a beverage is consumed that contains multiple transportable

carbohydrates such as glucose and fructose, gastrointestinal symptoms seem to be reduced compared with the consumption of the same amount of a single carbohydrate. Foods with high glycemic index and/or high fiber content may also contribute to osmotic “dumping,” whereas caffeine may produce diarrhea directly through its laxative properties.

Runner’s Diarrhea (“Runner’s Trots”)

Although all athletes may be at increased risk for diarrhea, runners are at particular risk for a specific running-related diarrheal condition often referred to colloquially as “runner’s trots.” This condition is classically encountered in distance runners, but its exact pathophysiology is still largely unknown. Conjectured mechanisms include impairment in GI tract function as a result of the repetitive jarring associated with lengthy stints of running, fluid and electrolyte shifts associated with prolonged exercise, and relative ischemia of the GI tract. It is clear that unique intrinsic factors also may play a role in this condition because some endurance athletes rarely suffer with it whereas others are chronically affected. Clinically, athletes experience an urgent need to defecate mid-run with stools that may be loose or essentially normal. Management focuses on prevention, as noted next.

Diarrhea Management

Management centers on training and lifestyle modification to allow the gut to perform optimally in the face of high levels of exertion. Modifying physical stress through reduction of training intensity, duration, and/or distance for 1 to 2 weeks may lead to symptom resolution.⁵ The athlete should then transition to cross-training for several weeks, followed by resumption of running at progressively higher levels. Adequate hydration is always a crucial part of lifestyle management, and avoidance of NSAIDs, aspirin, antibiotics, and caffeine may also be helpful. Dietary modification to include a low-residue, low-fiber diet with avoidance of sports drinks and/or gels with carbohydrate concentrations greater than 8% should be recommended. Avoiding solid food at least 2 hours prior to exercise may help, as may adherence to an antidiarrheal diet such as the BRAT (bananas, rice, applesauce, toast) diet. Making an effort to have a bowel movement before exercise can greatly minimize the likelihood of midactivity GI distress.

In general, anticholinergic medication use for exercise-associated diarrhea is discouraged because the side effect profile and physiologic properties of these medications have a negative impact on sweat rate and thermoregulation. Opiate- and atropine-based preparations should also be avoided. Loperamide (Imodium) may be considered in athletes with nonbloody diarrhea who are at risk for dehydration and associated heat illness but are otherwise clinically well. Similarly, antispasmodic agents such as dicyclomine or hyoscamine may be considered for severe symptoms, but their use must be weighed against the risk of anticholinergic side effects.

Stool studies should be obtained in athletes with “alarm” features, including gross hematochezia, profound diarrhea associated with dehydration, persistent diarrhea greater than 48 hours, fever, severe abdominal pain, recent travel, or possible exposure to infectious diarrheal pathogens. Consideration

should also be given to studies in immunocompromised patients or older athletes,⁴² especially if they have comorbid medical conditions.

Lower Gastrointestinal Bleeding

Microscopic heme-positive stools, rectal bleeding, or frank bloody diarrhea may be seen commonly with high-level training or competition particularly marathoning, ultramarathoning, or high-mileage cycling. Occult rectal bleeding has been demonstrated in approximately 25% of recreational triathletes, 20% of marathon runners, and almost 90% of 100-mile ultramarathoners.³⁴ More dramatically, frankly bloody stools have been reported in up to 6% of postmarathon runners.⁴³ As described earlier, long-duration events create prolonged ischemia in the GI tract, which compromises mucosal integrity, resulting in varying degrees of lower GI tract bleeding. This condition is generally benign and self-limited in the absence of underlying GI tract diseases. Pronounced or protracted rectal bleeding may predispose to anemia and may be a marker of a more worrisome underlying GI issue.

Ischemic Conditions

There are four distinct ischemic syndromes seen in the GI tract.⁹ Although primarily seen in patients with vascular comorbidities such as hypertension, diabetes mellitus, or tobacco abuse, they must be considered by the sports medicine practitioner in light of the breadth and variable health characteristics of the exercising population.

Colonic ischemia is the most common form of ischemic bowel disease. Crampy left lower quadrant abdominal pain and bowel movements of stool mixed with frank blood are the common presenting clinical features. Colonoscopy should be performed acutely in the setting of rectal bleeding to make a definitive diagnosis. Mucosal changes of ischemia, typically manifest by congestive and hemorrhagic vascular lesions, are the classic findings. This condition is self-limited and the associated colonic changes will typically resolve in 1 week, thus limiting the yield of diagnostic testing performed at that time or beyond.

The less common ischemic syndromes include acute arterial mesenteric infarction (AAMI), chronic arterial mesenteric ischemia (CAMI), and mesenteric venous thrombosis. Although a discussion of these entities is beyond the scope of this text, it is important to consider them for older athletes presenting with severe abdominal pain and lower GI bleeding.

Although some risk for GI ischemia is unavoidable in the most intense of activities, aggressive hydration and heat management remain the key means of attempting to prevent or minimize lower GI bleeding.

Irritable Bowel Syndrome

A functional GI disorder without evidence of structural, biomechanical, radiologic, or laboratory abnormalities, IBS affects up to 15% of the general population.⁴⁴ Two forms have been described—"diarrhea predominant" and "constipation predominant"—although the classic pattern typically involves alternating manifestations of both. IBS is twice as common in women and typically presents in the second or third decade of life.

Approximately 50% of patients who have IBS have comorbid psychiatric illnesses, most commonly depression and anxiety. Proposed mechanisms for IBS include increased GI sensitivity to various stimuli including stress, food, cholecystokinin secretion, impaired transit of bowel gas, visceral hypersensitivity, autonomic dysfunction, and altered immune activation.⁴⁵

The clinical presentation of IBS is variable, with the most common symptoms including cramping abdominal pain relieved by defecation, altered stool frequency, altered stool form (watery, mucus, loose, hard), altered stool passage (urgency, strain), a sense of incomplete evacuation, and abdominal distention especially following meals. Athletes with IBS rarely have nocturnal symptoms and do not manifest features of systemic illness or gross GI bleeding. The diagnosis is generally made clinically based on presentation and lack of evidence of other GI disorders. Current evidence does not support the routine use of laboratory studies or imaging to exclude organic GI disorders in patients who have typical IBS symptoms and are lacking "alarm" features such as significant abdominal pain, fevers, GI bleeding, or weight loss.⁴⁴

Primary management strategies focus on lifestyle measures including high-fiber diet, adequate hydration, and stress reduction techniques. For ongoing symptoms, drug treatment of IBS is based on the athletes' predominant symptoms. For diarrhea-predominant IBS, antidiarrheal (loperamide) or antispasmodic (dicyclomine, hyoscamine) medicines used as needed are typically efficacious. Constipation-predominant IBS is generally managed with an increase in the lifestyle measures noted previously coupled with nonstimulant laxatives such as polyethylene glycol (Miralax) or psyllium-containing products. Adequate hydration is particularly important in constipation-prone athletes.

Celiac Disease

Celiac disease is a hereditary autoimmune disorder resulting in malabsorption in the GI tract. Hypersensitivity of the immune system to gluten- or gliadin-containing foods is the primary pathophysiologic mechanism. Gluten protein found in wheat, barley, and rye results in an immunologic reaction in the intestinal mucosa that causes villous atrophy and subsequent impairment in absorption of important nutrients. Increasingly, celiac disease is being recognized and diagnosed in the general population and, as such, is also being more commonly encountered in athletes and active individuals. Although its primary pathology is found in the GI tract, celiac disease may also have effects on blood, bone, brain, nervous system, and skin.⁴⁶ Iron deficiency anemia has been reported in up to 70% of patients with newly diagnosed celiac disease,⁴⁷ and calcium and vitamin D deficiencies are also common, which may increase the risk of stress fractures and osteopenia.⁴⁸

The diagnosis of celiac disease is often delayed many months or even years because the presenting symptoms may be suggestive of other more common entities or may be sufficiently mild as to be largely ignored by the athlete. Signs and symptoms of classic celiac disease include chronic diarrhea, abdominal bloating/cramping/pain, malnutrition, fatigue, vomiting, anemia, myalgia/arthritis, osteopenia/osteoporosis, menstrual irregularities, irritability, constipation, short stature, and dermatitis

herpetiformis.⁴⁹ Initial serologic testing is traditionally via serum tissue transglutaminase (tTG) antibody levels. More specialized studies are available but are beyond the scope of our discussion. The “gold standard” for definitive diagnosis of celiac disease is tissue biopsy of small intestinal mucosa demonstrating classic blunted villi.

Eating a strict gluten-free diet is the only effective management of celiac disease, and this can be challenging on many levels for the competitive athlete. The elimination of all sources of wheat, rye, and barley requires that the athlete find ample alternative sources of carbohydrate to meet the recommended carbohydrate intake of 6 to 10 g/kg body weight. Beans, rice, corn meal, corn flour, nuts, potatoes, tapioca, and quinoa are excellent sources of gluten-free carbohydrates, along with fresh fruits and vegetables. Travel and team meals can be particular challenges for athletes with celiac disease who must ensure appropriate food choices to adequately meet their healthy fat, protein, and carbohydrate intake goals and to provide sufficient energy for maximum performance. In recent years, it has become increasingly popular for nonceliac athletes to consume a gluten-free diet because of perceived GI or general health benefits. However, research into the effects of gluten-free food intake in this group demonstrated no overall effect on performance, GI symptoms, well-being, indicators of intestinal injury, or inflammatory markers.⁵⁰

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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